

CSE-4102: IMAGE PROCESSING & COMPUTER GRAPHICS LABORATORY

PROJECT PROPOSAL

Title	Voyager 2: Journey Beyond the Solar System
Name	Sarwad Hasan Siddiqui
Roll	2107006

1. Project Idea

The project will be a small 3D OpenGL animation based on the journey of the Voyager 2 spacecraft. The scene will start near Earth and show Voyager 2 travelling past Jupiter, Saturn, Uranus and Neptune. After the Neptune encounter, the spacecraft will continue outward toward interstellar space. The purpose is to make a clear animated graphics scene where the objects and camera movements can demonstrate the transformation, animation, lighting and shading topics taught in CSE-4102.

2. Main Objects in the Scene

- Sun: central reference object and main light source.
- Earth: starting point of the Voyager 2 journey.
- Jupiter: first major flyby scene.
- Saturn: flyby scene with a visible ring system.
- Uranus and Neptune: later flyby scenes that extend the journey into the outer Solar System.
- Voyager 2: simplified 3D spacecraft made from basic OpenGL primitives such as boxes, cylinders and a dish antenna.
- Trajectory: a visible 3D path showing the spacecraft movement between the major mission points.
- Starfield and heliopause boundary: background and final-stage objects for a clearer space environment.

3. Animation and Transformation Features

- Voyager 2 will move along a predefined 3D trajectory using frame-based animation.
- The spacecraft will rotate to follow the direction of the trajectory.
- The planets will have slow rotation to make the scene active without distracting from the spacecraft.
- The trajectory will be drawn progressively so that the travelled path becomes visible during the animation.
- Curved sections near Jupiter, Saturn, Uranus and Neptune will visually represent gravity-assist flybys. No real-time gravity simulation is required.
- Translation, rotation, scaling and composite transformations will be used for the spacecraft, planets, rings and camera.
- Hierarchical transformations will be used for the Voyager model, so parts such as the antenna and instrument boom follow the main spacecraft transformation.

4. Mission Flow

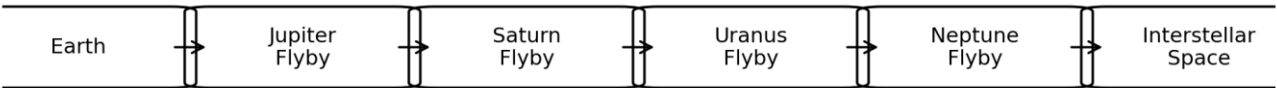


Figure 1. Main animation sequence

VOYAGER 2 PROJECT PROPOSAL

5. Lighting and Shading

- The Sun will be used as the main light source for the planetary scene.
- Ambient, diffuse and specular components will be used to produce different levels of illumination.
- Diffuse lighting will show how the visible side of each planet receives light.
- Specular lighting will be used on the Voyager spacecraft and selected surfaces to show highlights.
- The project will support Flat, Gouraud and Phong shading so the differences can be demonstrated during the presentation.
- Planet and spacecraft normals will be handled correctly for lighting calculations after transformation.

6. Camera and User Interaction

- Overview camera: shows the main planets and the full trajectory.
- Follow camera: stays near Voyager 2 while it travels between mission points.
- Flyby camera: focuses on the spacecraft and a selected planet during a close approach.
- Free camera: allows the user to inspect the scene from different directions.
- Keyboard controls will provide play/pause, restart, animation-speed control, camera switching, trajectory toggle and shading-mode selection.
- Mission labels will identify Earth, Jupiter, Saturn, Uranus, Neptune and the final interstellar stage.

7. OpenGL Implementation Plan

The application will be developed in C++ using OpenGL. GLFW can be used for the window and input system, GLAD for OpenGL function loading, and GLM for matrix and vector calculations. The main rendering loop will update the animation first and then render the scene using the current object transformations, camera, lights and shaders.

8. Important Graphics Concepts to be Demonstrated

- 3D translation, rotation and scaling of objects.
- Composite transformations and transformation order.
- Local and world coordinate systems.
- Hierarchical transformation of Voyager spacecraft parts.
- Camera/view transformation and multiple camera positions.
- Lighting with ambient, diffuse and specular components.
- Flat, Gouraud and Phong shading.
- Animation using time-based position updates and interpolation along a 3D trajectory.
- Depth testing, materials, surface normals and basic transparency for the heliopause boundary.

9. Extra Features

- Mission timeline showing the sequence from launch to interstellar space.
- On-screen information panel for each major planetary encounter.
- Adjustable animation speed for live demonstration.
- Progressive trajectory drawing and optional gravity-assist path markers.
- Saturn rings, starfield and a simple heliopause boundary to improve the final scene.

10. Scope and Expected Result

The project will use visually scaled planetary distances and a predefined trajectory so that the complete scene can be rendered clearly in a normal OpenGL window. It will not attempt to reproduce exact astronomical distances or perform a full gravitational simulation. The expected result is a short and interactive 3D animation in which Voyager 2 moves through the outer Solar System while the required transformation, animation, lighting and shading features are clearly visible.